multigrain _travelers guide _for navigating particle storms

8 sounds | **2** scenes per sound | **32** s per sample

`[ rec ]` press `[ rec ]` hold › then 1 s = keep holding `[ load ]` grey = alt label

home page unless marked · alt › = press alt first · knobs edit the blinking sound + scene

samples = .wav · 48_khz · 16-bit · stereo · ≤ 32_sec

red = hold

01 sounds

mgr_01

`[ 1–8 ]`	press / hold = play grains · active sound blinks
`[ latch ]` › `[ 1–8 ]`	**latch** it on · switching keeps it latched
`[ latch ]` + `[ 1–8 ]`	select silently, without playing
`[ 1–8 ]` + `[ 1–8 ]`	**copy sound** to an empty sound
`[ latch ]` + `[ clear ]`	**kill all grains** (blur tails on)
`[ gate ]` `[ next ]`	play the sound · step one right (normalled)
`[ select ]`	0–5 v picks 1–8 · empties skipped

02 scenes · morph

mgr_02

`[ scene a ]` `[ b ]`	which scene the knobs edit (blinks)
fader	a ← → b · you always hear the fader
`[ scene ]` + `[ scene ]` / `[ 1–8 ]`	**copy scene** to the other one or a sound
`[ rand ]` / `[ clear ]` + `[ scene ]`	randomize / reset that scene
`[ morph ]` cv	±5 v = full sweep · knob attenuverts
shape · reverse	don't blend: each grain picks a or b by chance

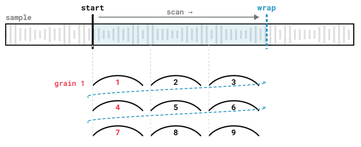
03 position

mgr_03

`[ start ]`	where grains begin in the sample
`[ wrap ]`	how far scan travels before wrapping
`[ scan ]`	noon = still · cw forward · ccw back
resets	sound press or gate = back to start
live sounds	start cw = the newest audio coming in
`[ reverse ]` long	**ping-pong**: bounce, don't wrap (fades)

__fig.01 : the scan

start · wrap · scan



reaching wrap sends the next grain back to start

noon = every grain from start · ccw scans backward · ping-pong bounces

04 grain

mgr_04

`[ rate ]`	0 hz (one grain per trigger) – 130.8 hz
`[ size ]`	20 ms – 4 s · noon ≈ 1 s
`[ pitch ]`	±2 oct (±3 with mod) · green = octaves
`[ tone ]`	ccw lowpass · noon open · cw highpass
`[ level ]` `[ blur ]`	grain volume · reverb send + decay
`[ reverse ]`	grains play backwards (per scene)



tukey = default · discrete: morph picks per grain by chance

05 links · sync

mgr_05

`[ rate–size ]`	size sets the rate · rate = 1× – 8×
`[ size–pitch ]`	size = audio amount: octave up = half
alt › `[ sync ]`	rate ×/÷ 2 4 8 16 of clock · noon ×1
sync + `[ rate–size ]`	size snaps to clock divisions



rate–size linked: size sets the base rate, rate multiplies it

07 clear · control all

mgr_07

`[ clear ]` › item ×2	**safe clear** · red = clearable
`[ clear ]` + item	quick clear, no confirm
items	sound (×1 settings, ×2 sample) · scene · mod · sample memory · load = init · quant
`[ scene ]` + turn	value goes to both scenes
`[ latch ]` + turn	every sound · + scene = everything
`[ rand ]` + `[ 1–8 ]`	randomize a whole sound

12 quantizer

mgr_12

alt › `[ quant ]` / `[ quant ]`	open it (hold quant from home)
keys	clear rand x y z black · 1–7 white
notes	root yellow · scale purple · hold = root
`[ quant ]`	knob + mod (dim) / mod only (lit)
`[ latch ]` + note	apply to every sound

06 modulation

mgr_06

`[ rand ]` `[ x ]` `[ y ]` `[ z ]`	mod page · hold = momentary
mod page	knobs attenuvert: cw + yellow, ccw – purple
`[ rand ]` / `[ x ]` + turn	quick assign from home
`[ x ]` `[ y ]` `[ z ]` jacks	±5 v = full range · pitch 100% = 1v/oct
`[ rand ]` + `[ x ]` / `[ alt ]`	randomize x's (or rand's) assigns
`[ clear ]` › turn	on a mod page: clear that knob
rand · start	cw spray · ccw = anywhere in wrap
rand · rate	ccw = trigger chance 100% → 3%
rand · level, tone	ccw = random stereo (l / r opposite)

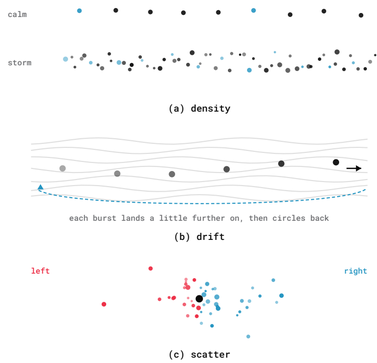
13 sample loader

mgr_13

alt › `[ /project ]` `[ /wavs ]` `[ /recs ]`	load from that folder
`[ ‹prev ]` `[ next› ]`	step · hold = 5 at a time
`[ 1–8 ]`	preview through the sound's grains

__fig.02 : the storm

mm_04



08 sampling

mgr_08

`[ sample ]`	rec (red) · again = stop · 32 s max
then	blue sample = save · empty 1–8 = assign
`[ sample ]` 1 s	grab the last 32 s · or open trim
trim page	start / end · scene a / b preview · dir · scan = nudge 25–250 ms
`[ alt ]` + `[ sample ]`	settings: gain 0.5–8.5× · thrsh · norm
`[ thrsh ]`	rec waits for the input (sample fades red)
`[ resamp ]`	inputs · resample outs · looping (magenta)

09 live · looping

mgr_09

`[ 1–8 ]` + `[ sample ]`	**live sound** from the inputs (magenta)
`[ latch ]` + `[ 1–8 ]` + `[ sample ]`	same, no trigger
`[ sample ]`	magenta = recording · cyan = frozen
`[ sample ]` 1 s	save the loop · 8–32 s memory
`[ thrsh ]`	auto-freezes when the input goes quiet

10 alt page

mgr_10

`[ alt ]` / `[ alt ]`	grey labels · hold = momentary
alt › `[ pitch ]`	fine-tune ±7 semitones
alt › `[ a ]` `[ b ]`	cycle output modes (see colours)
`[ 1–8 ]` on alt	sample live · load choke · save gate
`[ 1–8 ]` + `[ save ]` 1 s	save default sound (load = reset)

11 presets · projects

mgr_11

alt › `[ load ]` `[ save ]` `[ project ]`	preset / project pages
`[ x ]` `[ y ]` `[ z ]` › `[ 1–8 ]`	bank (twice = 4–6) › slot
`[ save ]` / `[ load ]`	quick save · quick reload
`[ 1–8 ]` 4 s	lock preset · collect project
`[ clear ]` + `[ load ]`	fresh empty preset
room	48 projects × 48 presets × 8 sounds

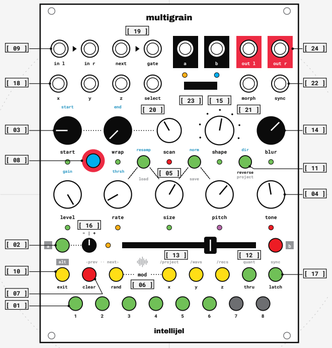
pdf print

__fig.03 : front panel

01–13 = sections · 14–24 = hardware

`[ 01 ]` __1–8 : the eight sounds
`[ 02 ]` __scene a / b : scenes, fader
`[ 03 ]` __start wrap scan : position
`[ 04 ]` __grain knobs : six grey knobs
`[ 05 ]` __links : rate–size, size–pitch
`[ 06 ]` __mod : rand · x · y · z
`[ 07 ]` __clear : clear · reset
`[ 08 ]` __sample : record · freeze
`[ 09 ]` __in 1 / r : audio in · live
`[ 10 ]` __alt / exit : alt page
`[ 11 ]` __reverse : reverse · project
`[ 12 ]` __quant : quantizer
`[ 13 ]` __alt › x y z : sample loader

`[ 14 ]` __blur : reverb
`[ 15 ]` __shape : 7 grain windows
`[ 16 ]` __morph att : cv attenuverter
`[ 17 ]` __thru / latch : monitor · latch
`[ 18 ]` __x y z : cv in ±5 v
`[ 19 ]` __next / gate : step · play
`[ 20 ]` __select : 0–5 v sound pick
`[ 21 ]` __morph : fader cv ±5 v
`[ 22 ]` __sync : clock in
`[ 23 ]` __a / b : cv outs + microsd
`[ 24 ]` __out l / r : audio out



_where things hide

25 · mgr_25

system settings

alt › hold project 4 s

- `rand` random: unipolar (default) / bipolar
- `latch` next jack = scan reset
- `x` auto-collect on preset save
- `start` gate delay, up to 10 ms
- `wrap` select span 5 v › 0.667 v (keyboard)
- `size` max grain: 4 s · ≤ 4 s / sample · ≤ 8 s · full
- `rate–size` shrink grains at high rates
- `clear` 30 s idle = back home (on)
- `thru` knobs jump (default) / catch
- `pitch` a / b 1v/oct trim · size–pitch = test notes

button LEDs

- `lit` on / available
- `dim` off, but can be turned on
- `off` not available here
- `blinking` selected (active sound / scene)
- `rapid` that page is open
- `fading` leads to a page or an action

microsd · boot

- `card` fat32 · 32 gb card included
- `folders` Project01–48 · /Wavs · /Recs
- `latch + clear` hold 4 s = reboot
- `at boot` clear = reset globals · sample / load / save / project = LED theme

LED colours

default theme · no screen, so colour is the display

`1–8`	loaded · playing · missing · live · live playing
`sample`	listening · recording · looping · frozen
`mod · pitch`	+ mod · – mod · quantized · octave · between
`a / b outs`	morph · grain trig · grain env · in env · out env · random · pitch · rec · in gate
`firmware`	version, shown at boot